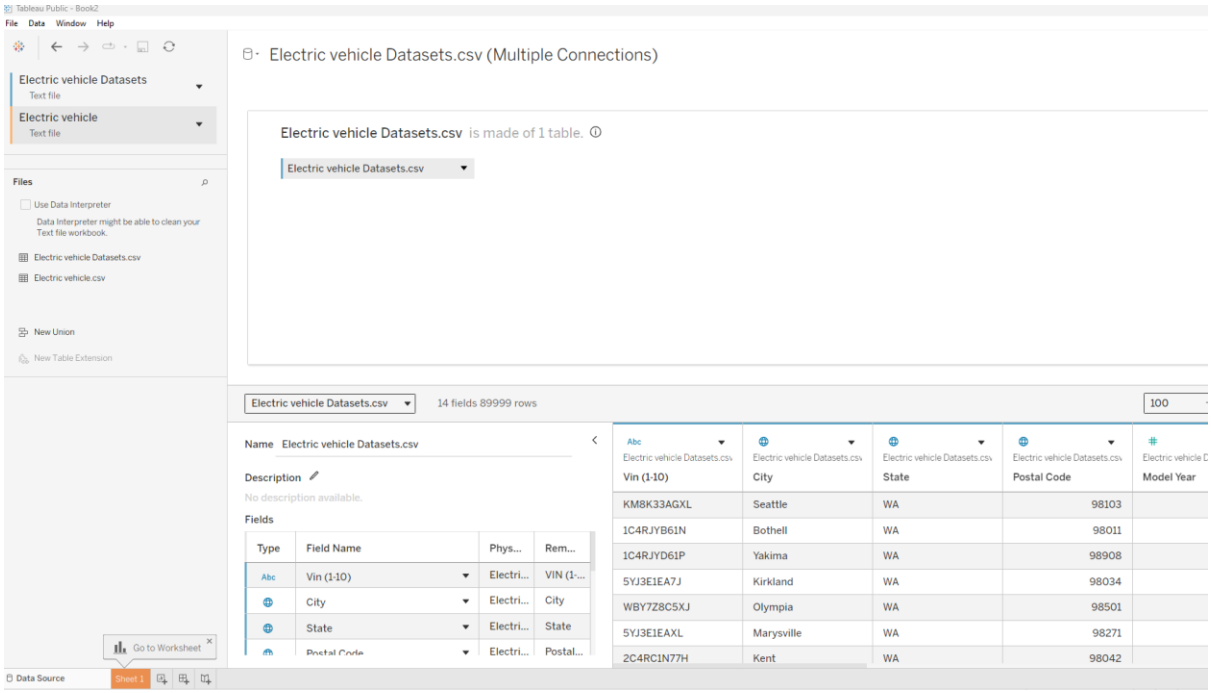
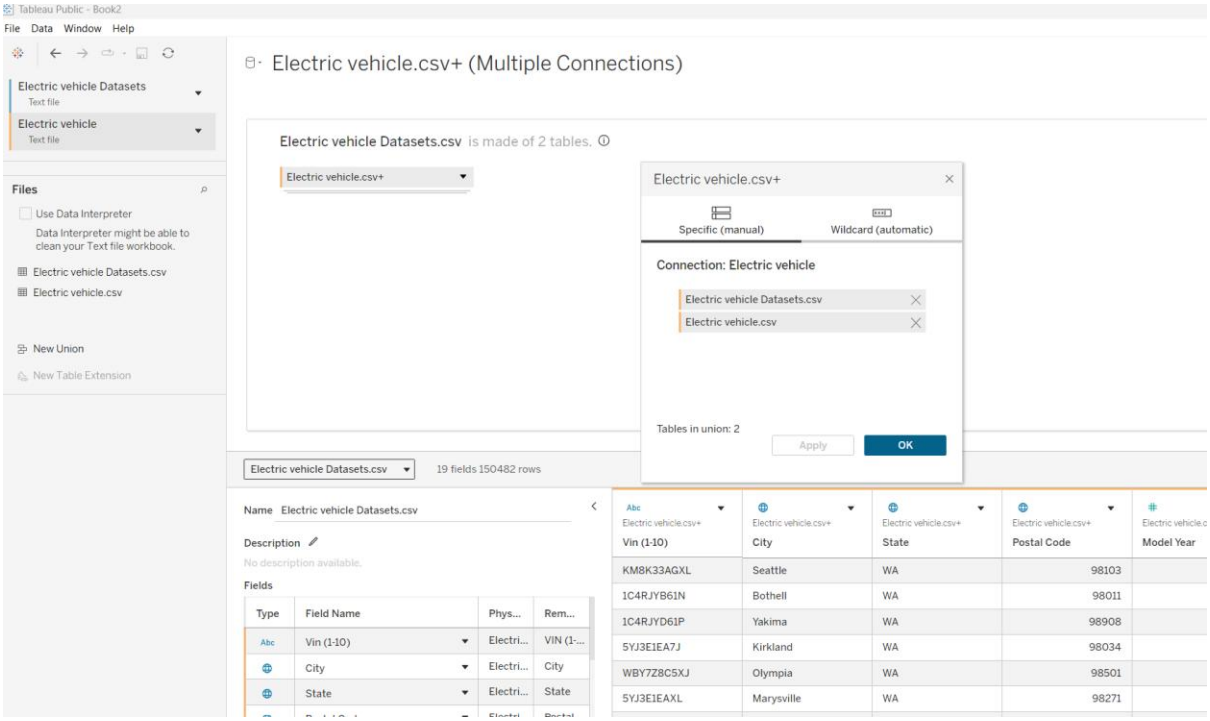
ASSIGNMENT 5 – TABLEAU

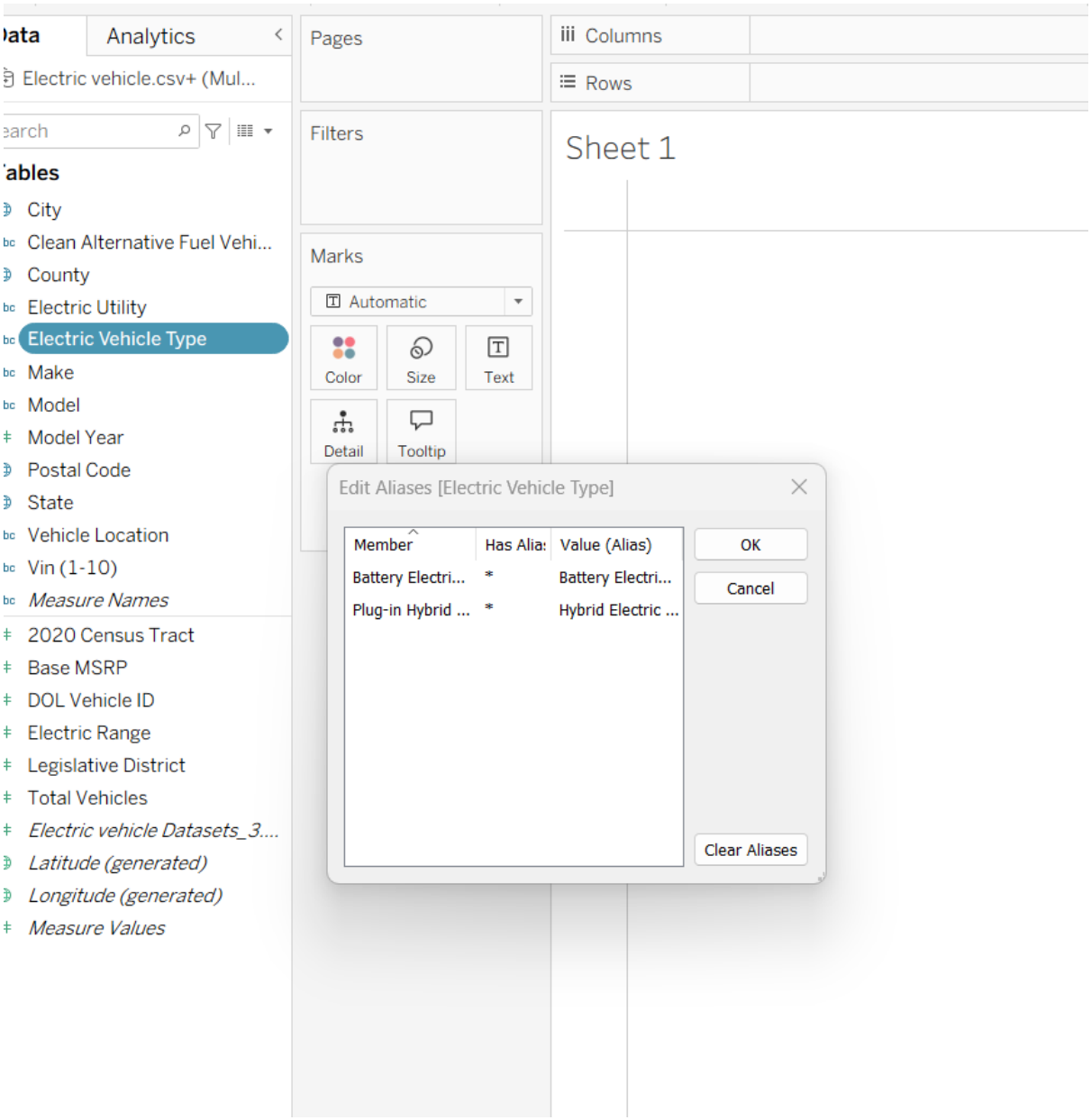
1. Import the 2 CSV files attached in this document in Tableau and combine them in a single file in Tableau.



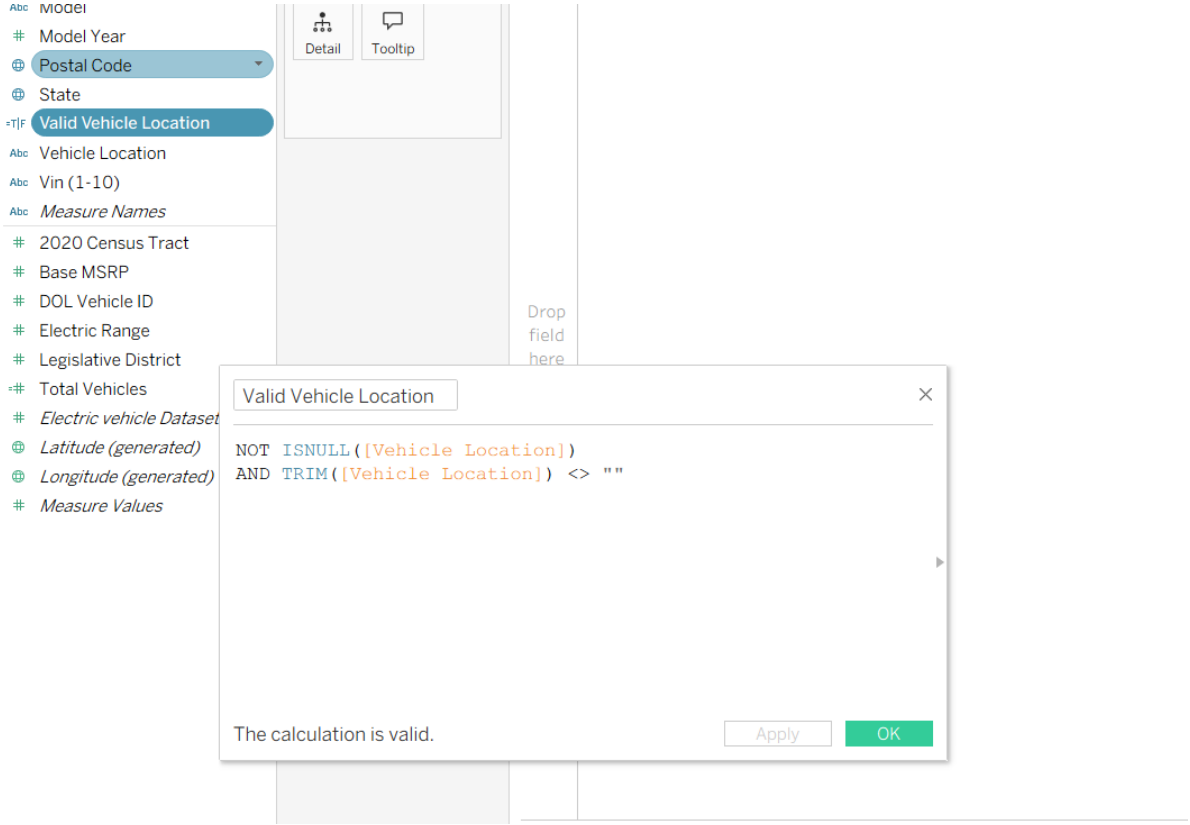


2. Perform the following data cleaning operations in Tableau.

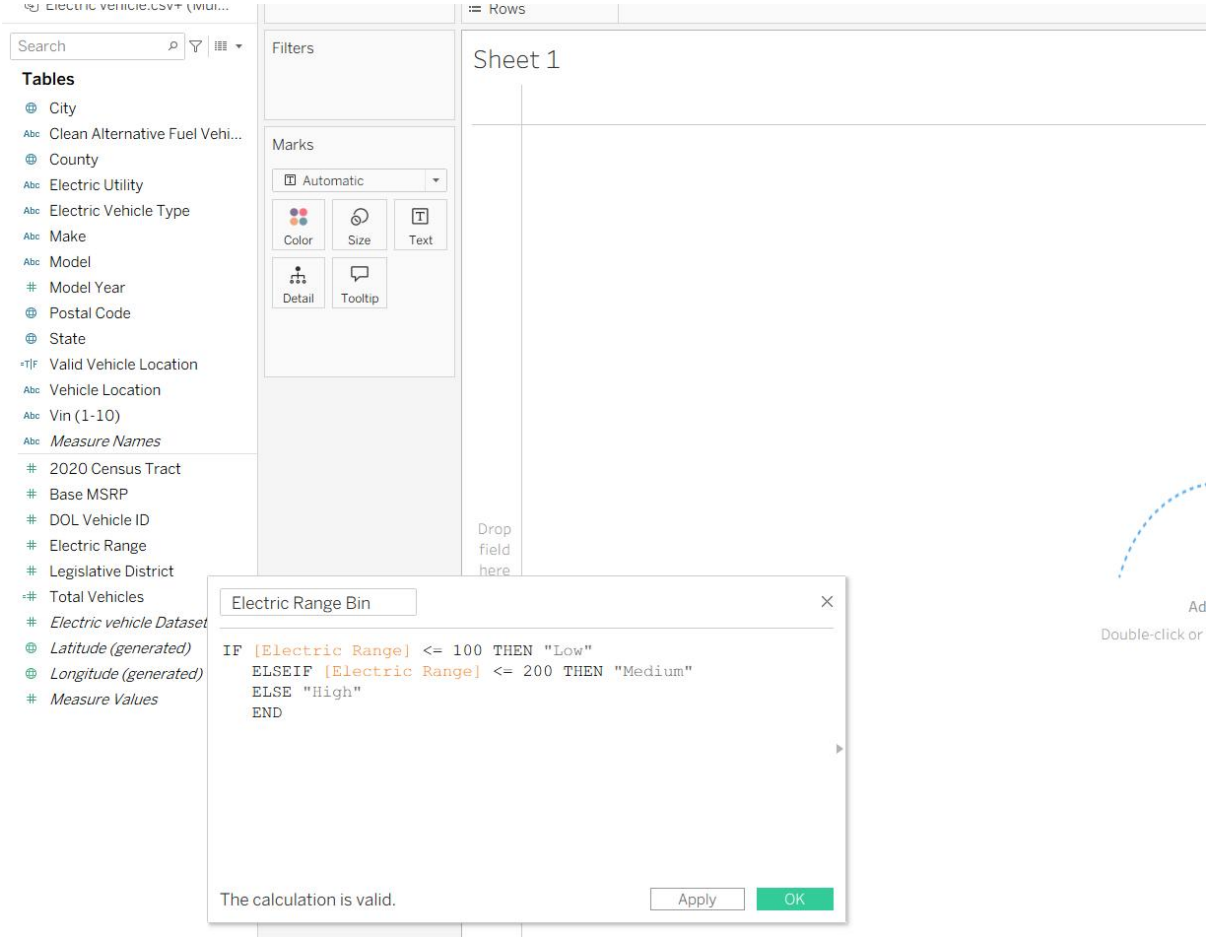
    a. Rename the values of Electric vehicle type column. Battery Electric Vehicle (BEV) to Battery Electric Vehicle. Plug-in Hybrid Electric Vehicle (PHEV) to Hybrid Electric Vehicle.



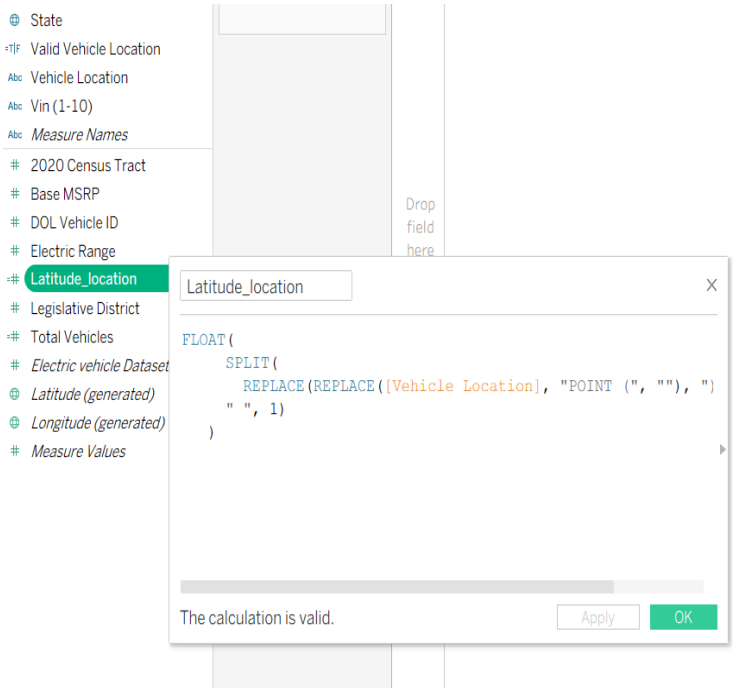
b. Exclude all the records from all the analysis where Vehicle Location column is Blank

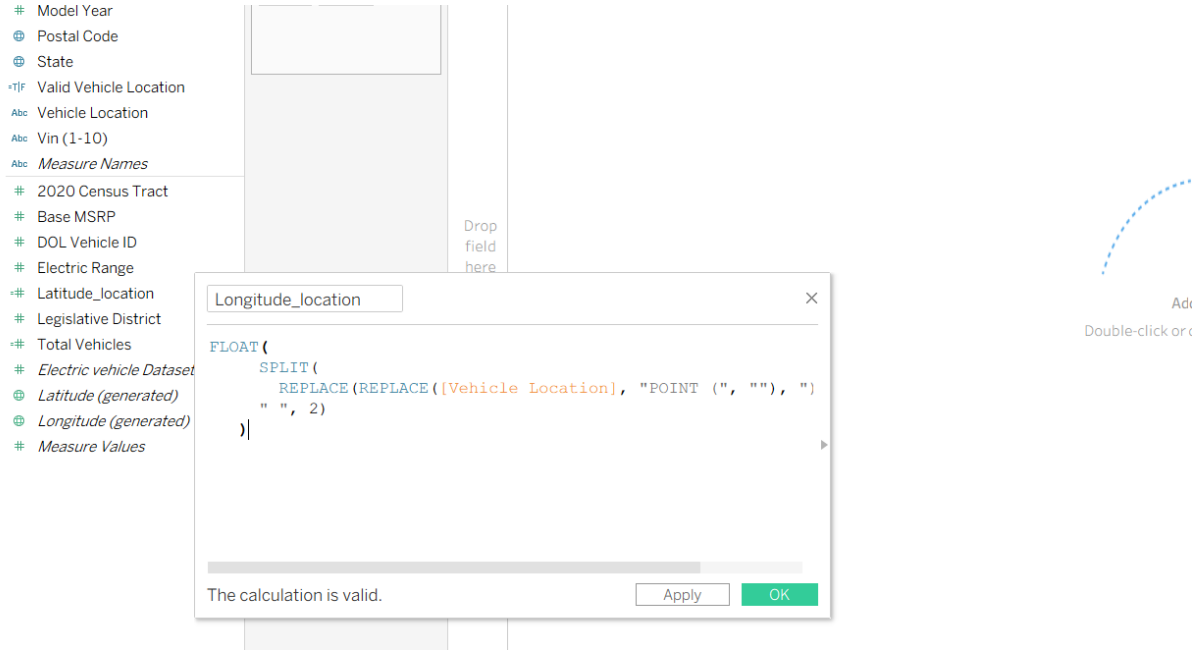


c. Create a new variable Electric Range Bin which will have 3 values (High, low or Medium). If electric range column has value 0-100 then value will be low, if 101-200 then medium, if greater than 200 then High.



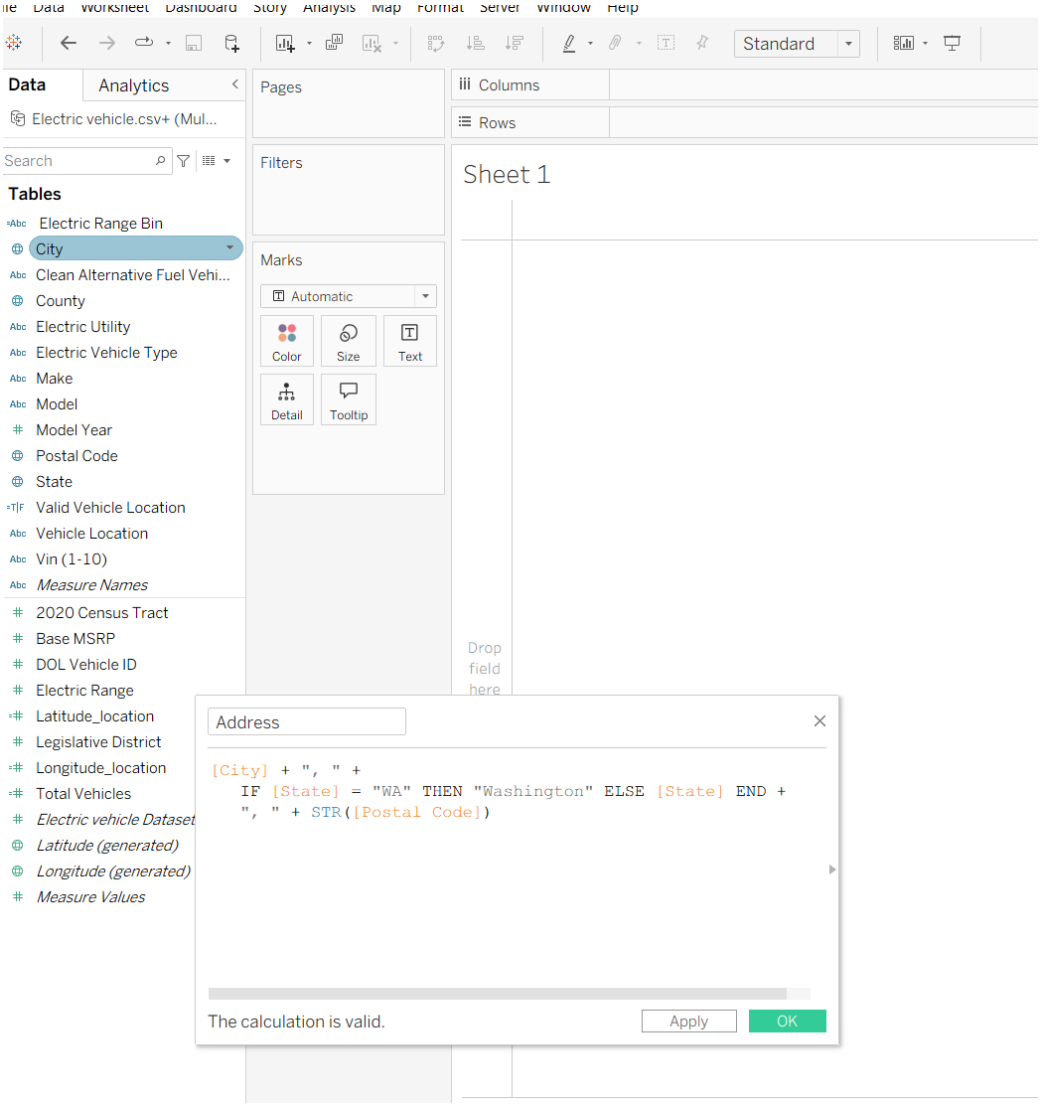
d. In vehicle location column, latitudes and longitudes of the locations are given. Create 2 new columns based on this column. First column name will be Latitude_location which will have first numeric value and 2nd column will be Longitude_location which will have second numeric value. Eg: POINT (-122.34301 47.659185) will have -122.34301 in latitude_location and 47.659185 in longitude_location.





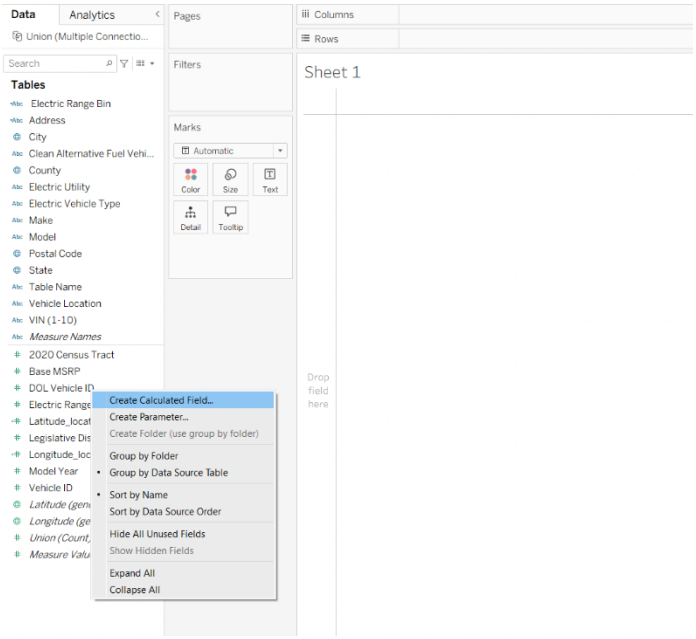
e. Create a new column(Address) using city, state and postal code. New column name will be combinations of those 3 columns - City,State, Postal code. Also, If state is WA, replace it with Washington.

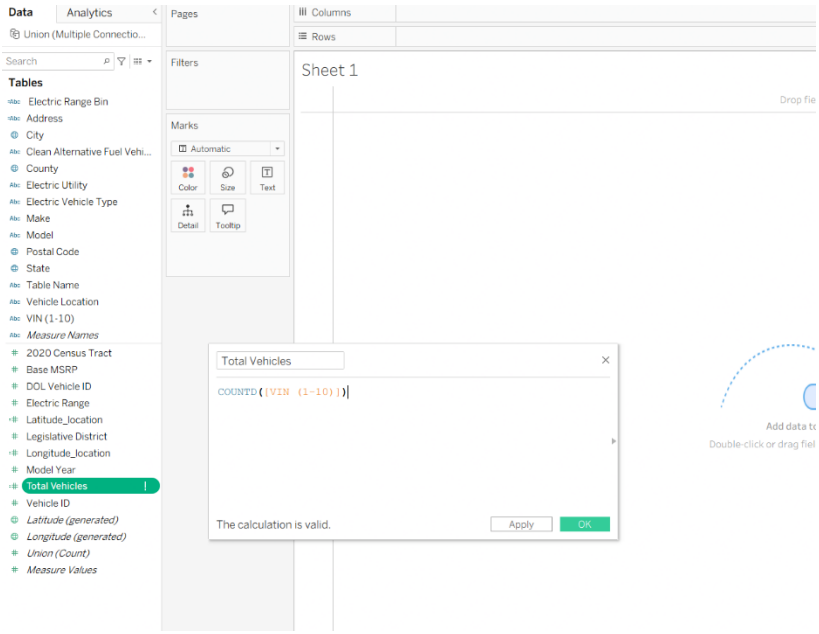
Eg: Seattle, Washington, 98103.

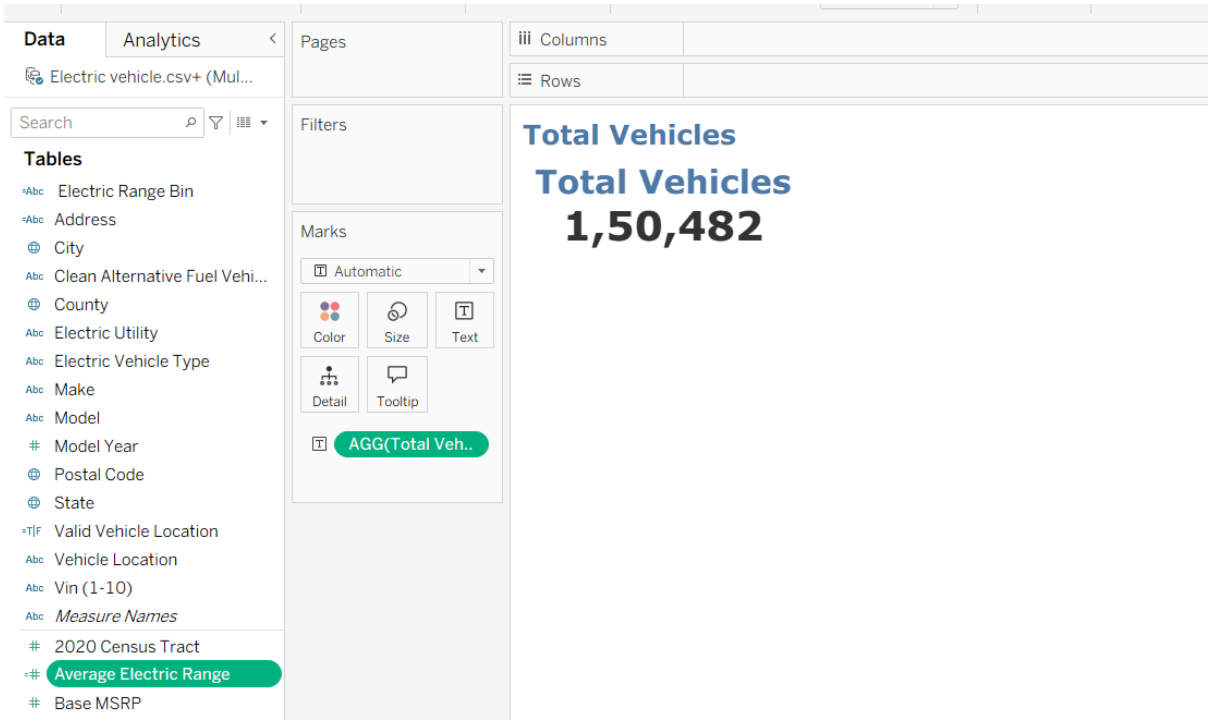


3. **KPIs** requirement for Dashboard build:

   a) Total Vehicles:
   Understand the overall landscape of electric vehicles, encompassing both BEVs and PHEVs, to assess the market's size and growth.

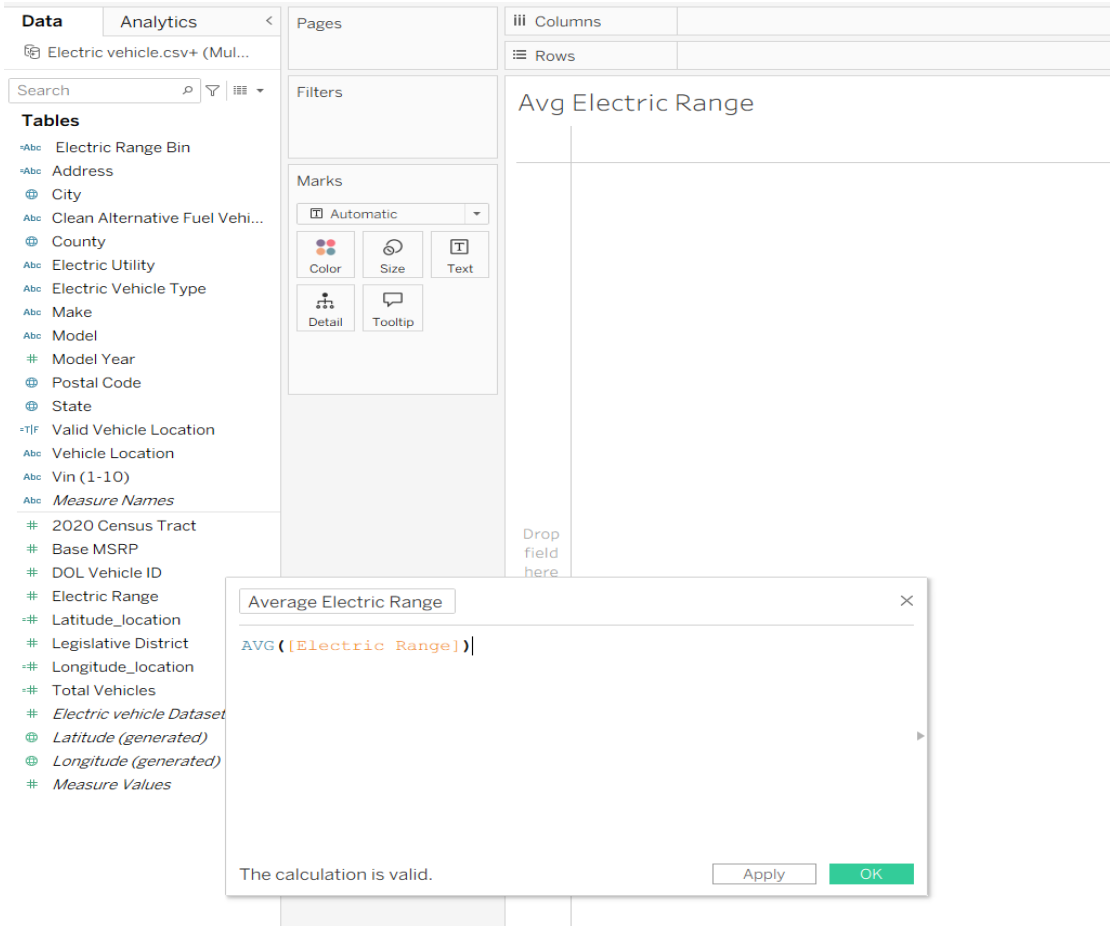

Data
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Tables
Electric Range Bin
Address
City
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County
Electric Utility
Electric Vehicle Type
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Postal Code
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VIN (1-10)
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Sheet 1
Drop field here
Create Calculated Field...
Create Parameter...
Create Folder (use group by folder)
Group by Folder
Group by Data Source Table
Sort by Name
Sort by Data Source Order
Hide All Unused Fields
Show Hidden Fields
Expand All
Collapse All

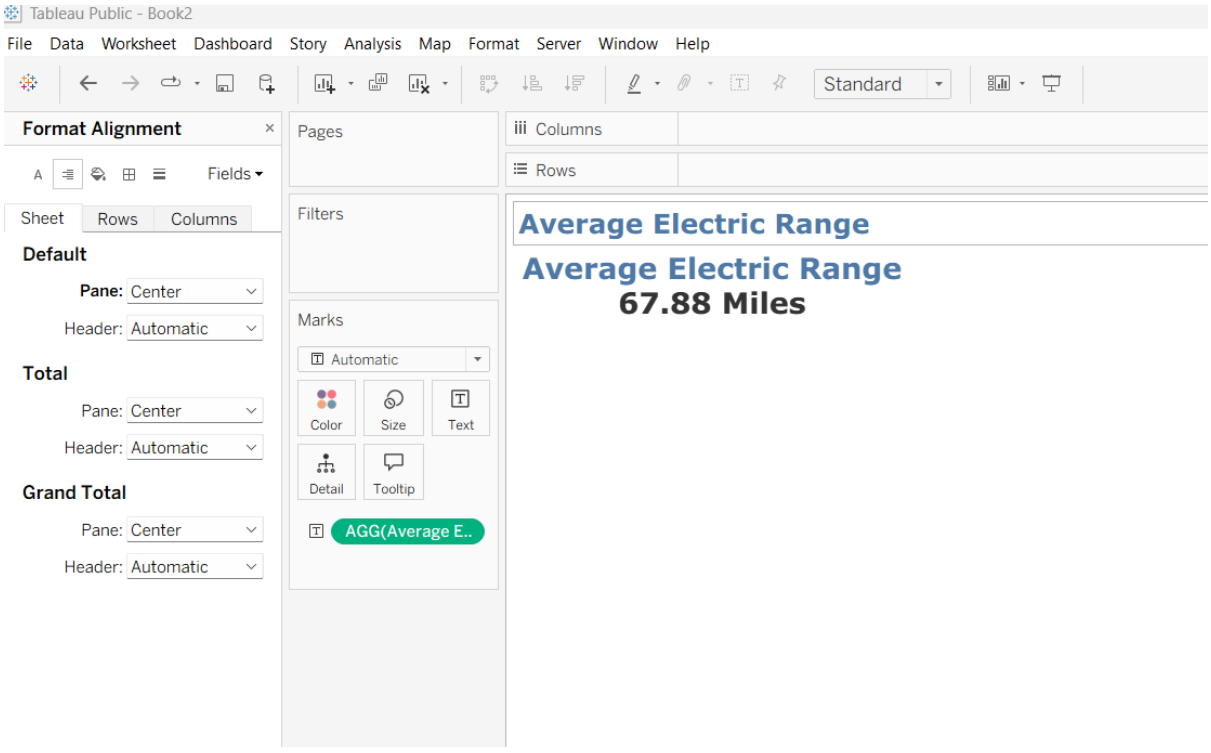

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County
Electric Utility
Electric Vehicle Type
Make
Model
Postal Code
State
Table Name
Vehicle Location
VIN (1-10)
Measure Names
2020 Census Tract
Base MSRP
DOL Vehicle ID
Electric Range
Latitude_location
Legislative District
Longitude_location
Model Year
Total Vehicles
Vehicle ID
Latitude (generated)
Longitude (generated)
Union (Count)
Measure Values
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Filters
Marks
Automatic
Color
Size
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Tooltip
Columns
Rows
Sheet 1
Drop fie
Total Vehicles
COUNTD([VIN (1-10)])
The calculation is valid.
Apply
OK
Add data to
Double-click or drag fiel



b) Average Electric Range:

Determine the average electric range of the electric vehicles in the dataset to gauge the technological advancements and efficiency of the EVs.

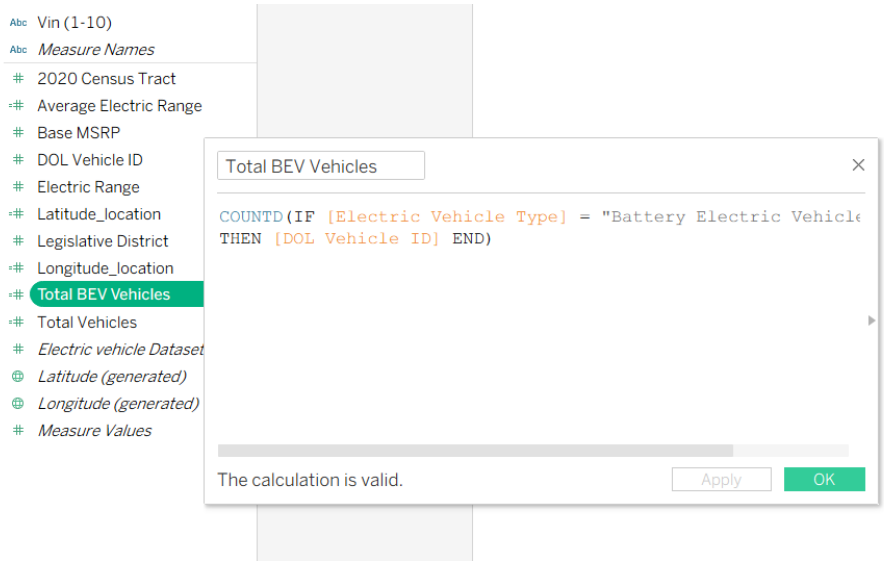

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County
Electric Utility
Electric Vehicle Type
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Model
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2020 Census Tract
Base MSRP
DOL Vehicle ID
Electric Range
Latitude_location
Legislative District
Longitude_location
Total Vehicles
Electric vehicle Dataset
Latitude (generated)
Longitude (generated)
Measure Values
Pages
Filters
Marks
Automatic
Color
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Detail
Tooltip
Columns
Rows
Avg Electric Range
Drop field here
Average Electric Range
AVG([Electric Range])
The calculation is valid.
Apply
OK

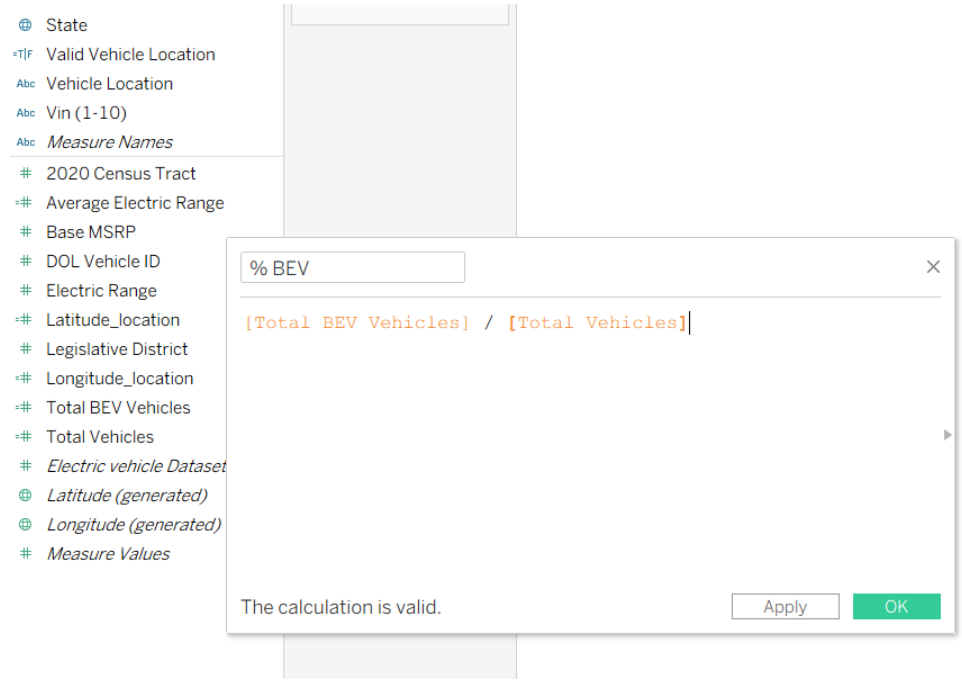

Tableau Public - Book2
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Standard
Format Alignment
Fields
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Default
Pane: Center
Header: Automatic
Total
Pane: Center
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Pane: Center
Header: Automatic
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Columns
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Average Electric Range
Average Electric Range
67.88 Miles

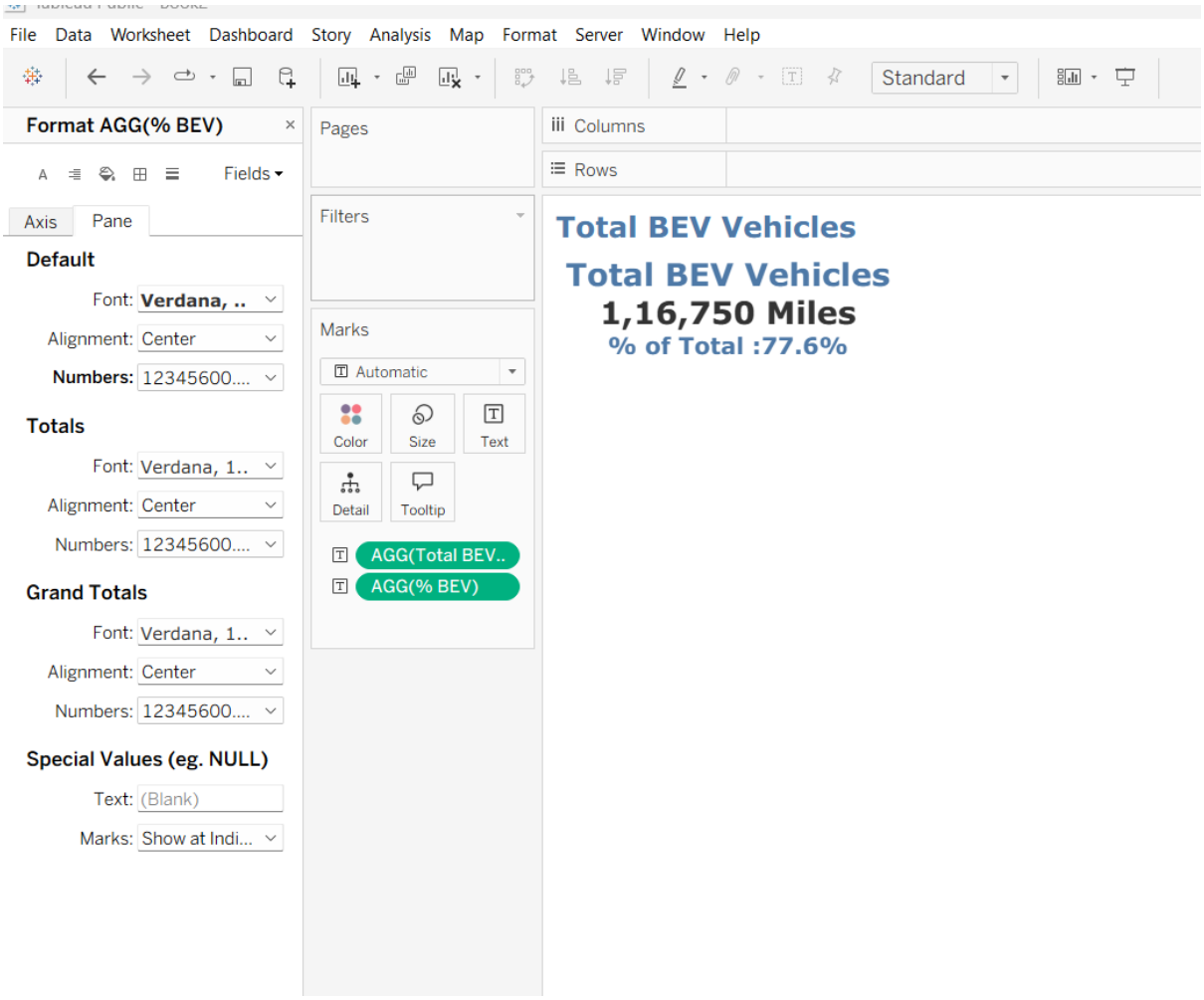
c) Total BEV Vehicles and % of Total BEV Vehicles:

Identify and analyse the total number of Battery Electric Vehicles (BEVs) in the dataset.

Calculate the percentage of BEVs relative to the total number of electric vehicles, providing insights into the dominance of fully electric models.



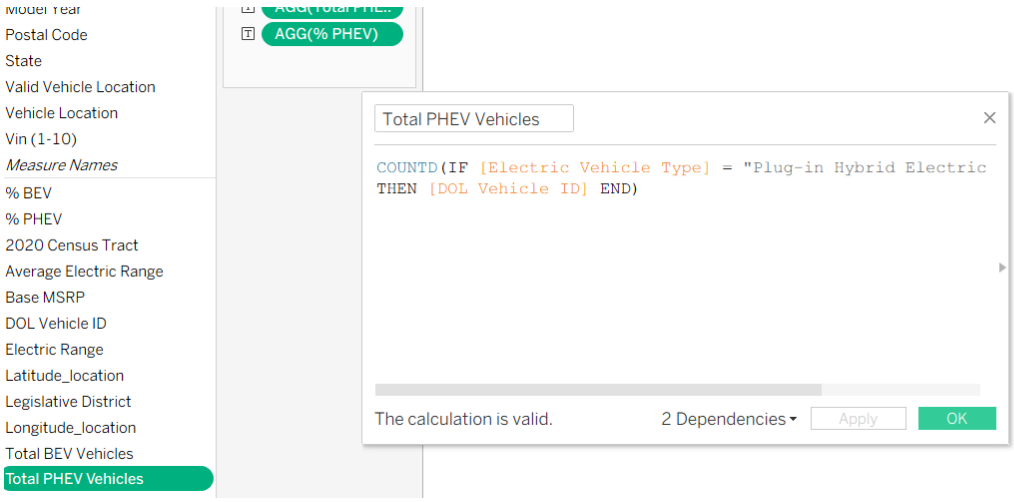

State
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2020 Census Tract
Average Electric Range
Base MSRP
DOL Vehicle ID
Electric Range
Latitude_location
Legislative District
Longitude_location
Total BEV Vehicles
Total Vehicles
Electric vehicle Dataset
Latitude (generated)
Longitude (generated)
Measure Values
% BEV
[Total BEV Vehicles] / [Total Vehicles]
The calculation is valid.
Apply
OK

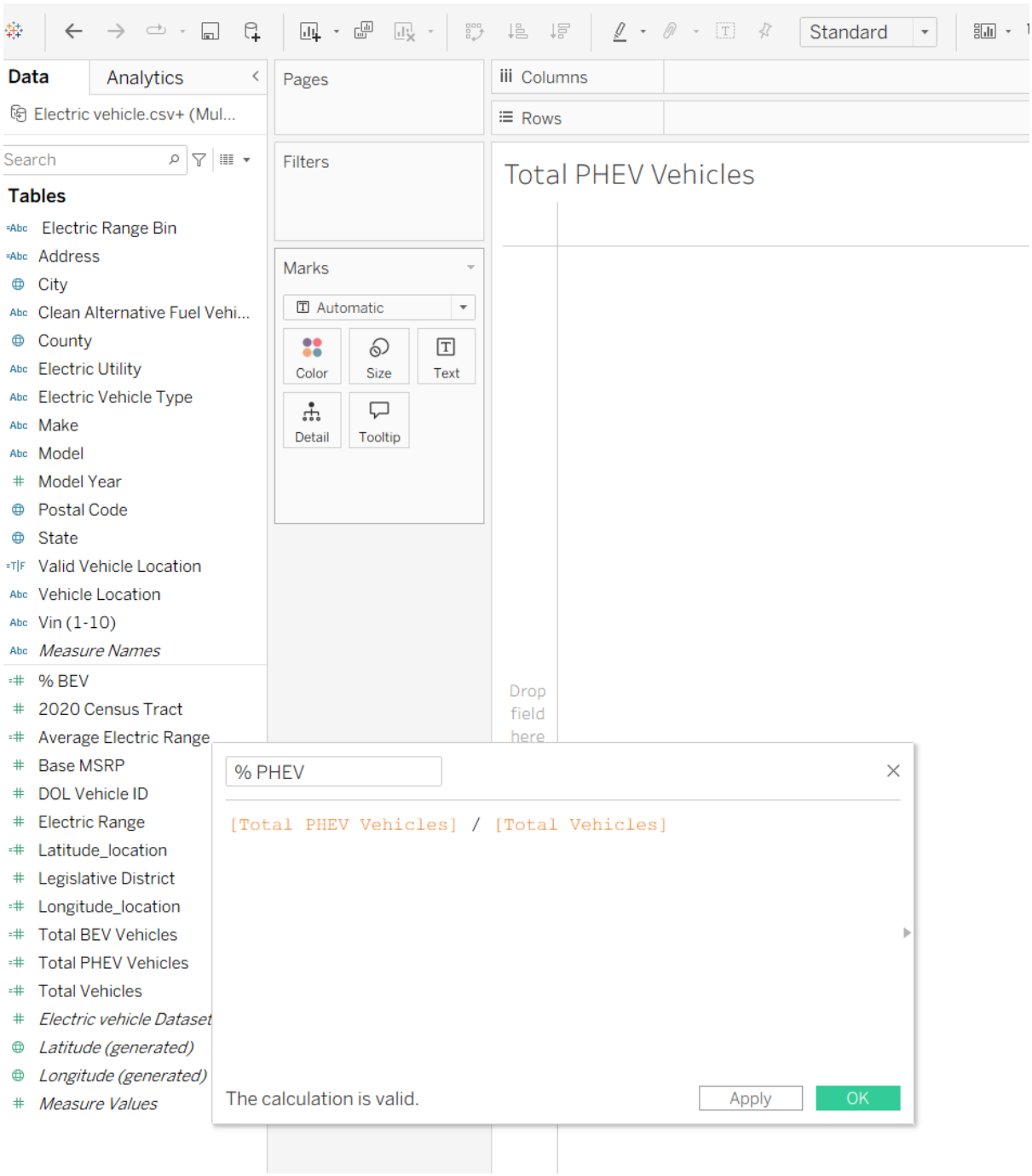

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
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Alignment: Center
Numbers: 12345600....
Grand Totals
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Special Values (eg. NULL)
Text: (Blank)
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AGG(Total BEV..
AGG(% BEV)
Columns
Rows
Total BEV Vehicles
Total BEV Vehicles
1,16,750 Miles
% of Total :77.6%

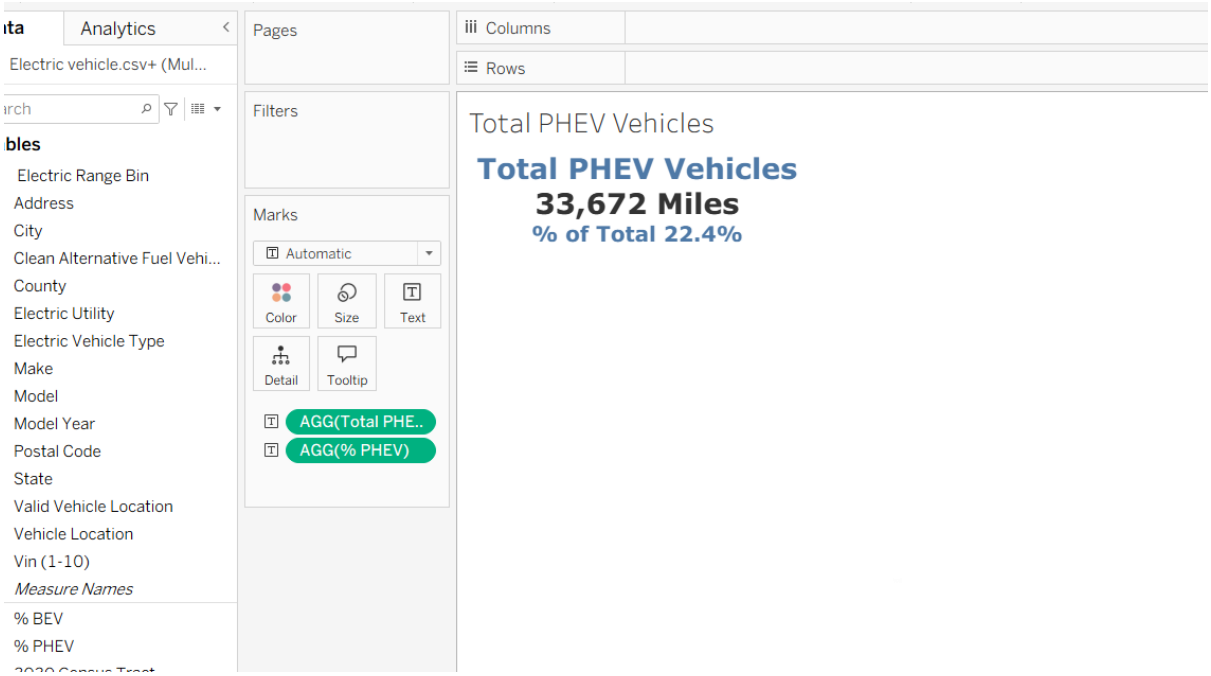
d) Total PHEV Vehicles and % of Total PHEV Vehicles:

Identify and analyze the total number of Plug-in Hybrid Electric Vehicles (PHEVs) in the dataset.

Calculate the percentage of PHEVs relative to the total number of electric vehicles, offering insights into the market share of plug-in hybrid models.

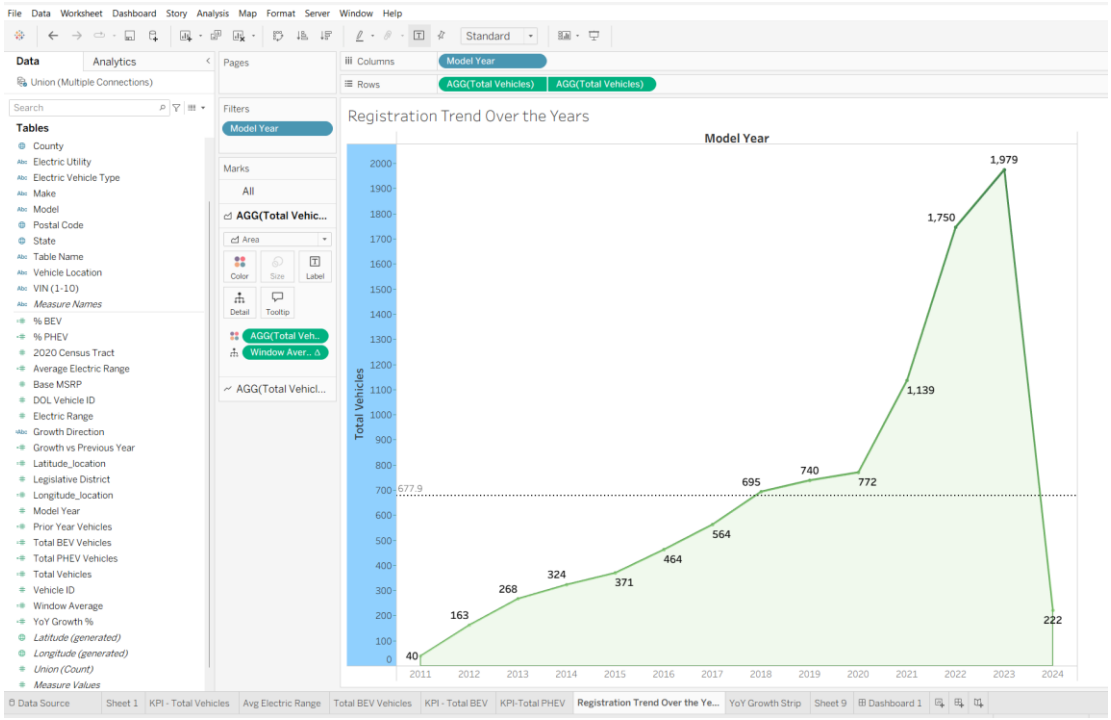



Standard
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Electric Range Bin
Address
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Clean Alternative Fuel Vehi...
County
Electric Utility
Electric Vehicle Type
Make
Model
Model Year
Postal Code
State
Valid Vehicle Location
Vehicle Location
Vin (1-10)
Measure Names
% BEV
2020 Census Tract
Average Electric Range
Base MSRP
DOL Vehicle ID
Electric Range
Latitude_location
Legislative District
Longitude_location
Total BEV Vehicles
Total PHEV Vehicles
Total Vehicles
Electric vehicle Dataset
Latitude (generated)
Longitude (generated)
Measure Values
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Marks
Automatic
Color
Size
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Detail
Tooltip
Columns
Rows
Total PHEV Vehicles
Drop field here
% PHEV
[Total PHEV Vehicles] / [Total Vehicles]
The calculation is valid.
Apply
OK



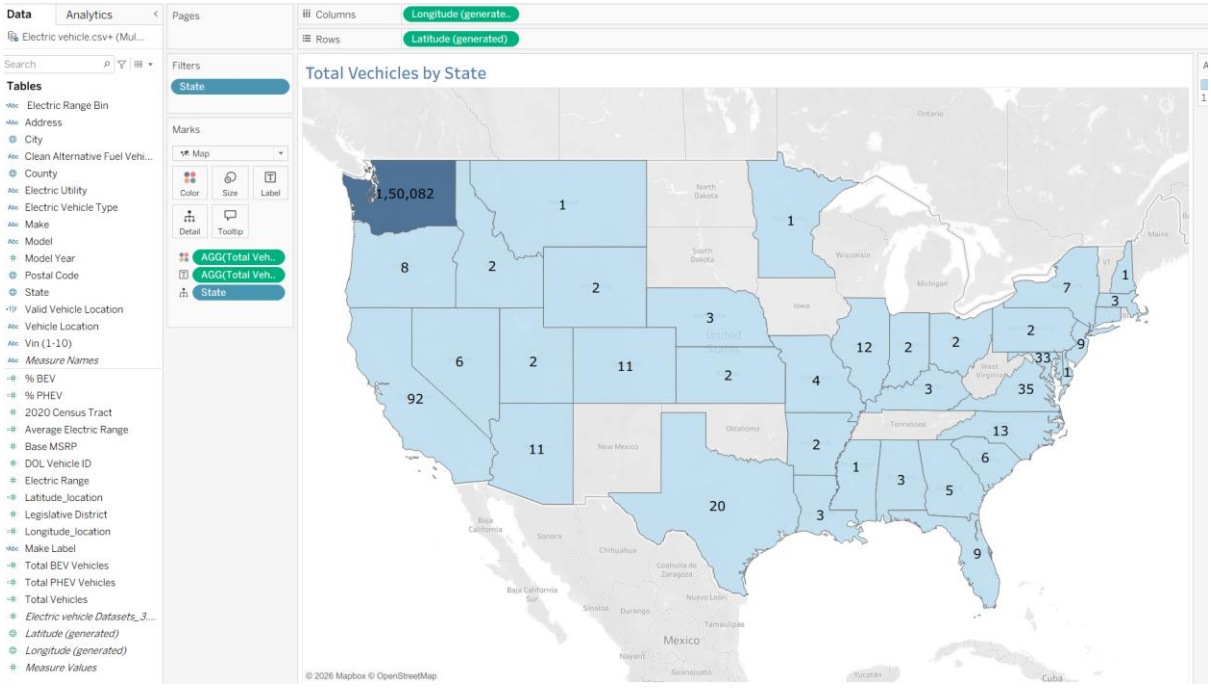
4. **Charts** Requirement:

    a) Total Vehicles by Model Year (From 2011 Onwards):
    Description: This chart will illustrate the distribution of electric vehicles over the years, starting from 2011, providing insights into the growth pattern and adoption trends.

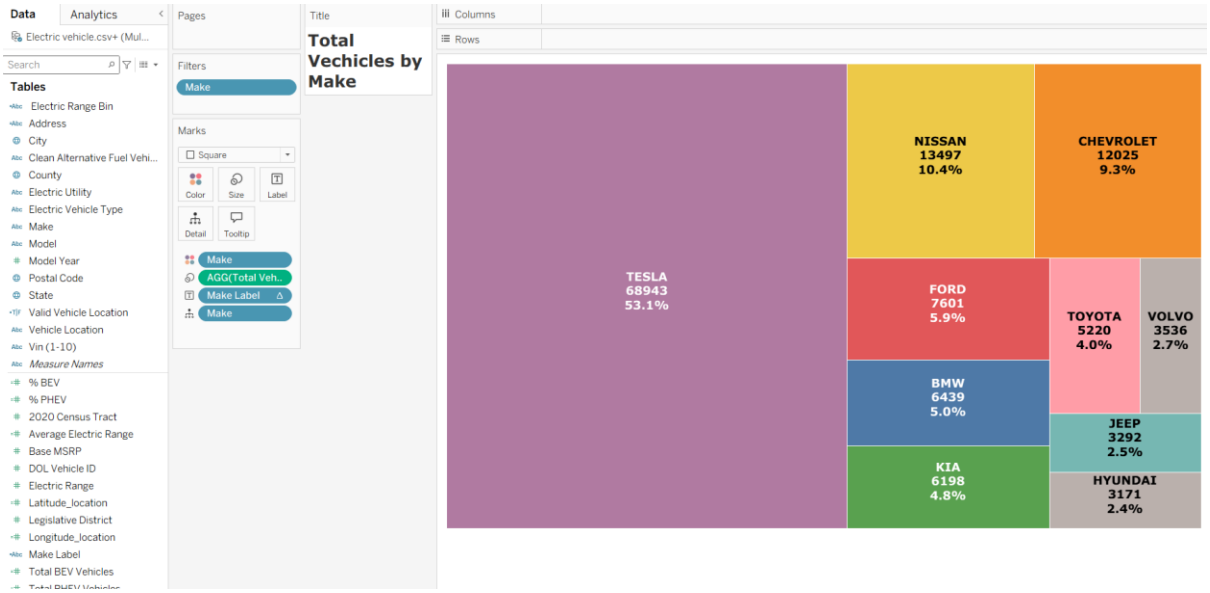


b) Total Vehicles by State:

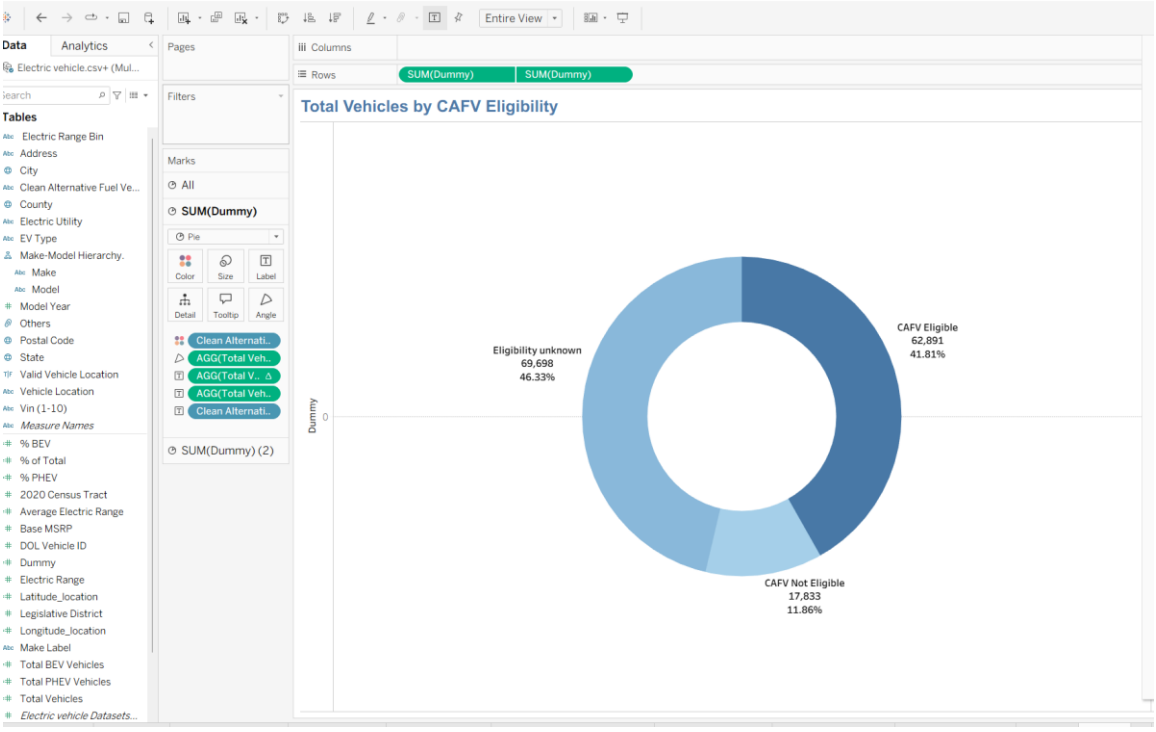
Description: This chart will showcase the geographical distribution of electric vehicles across different states, allowing for the identification of regions with higher adoption rates.



c) Top 10 Total Vehicles by Make:
Description: Highlight the top 10 electric vehicle manufacturers based on the total number of vehicles, providing insights into the market dominance of specific brands.

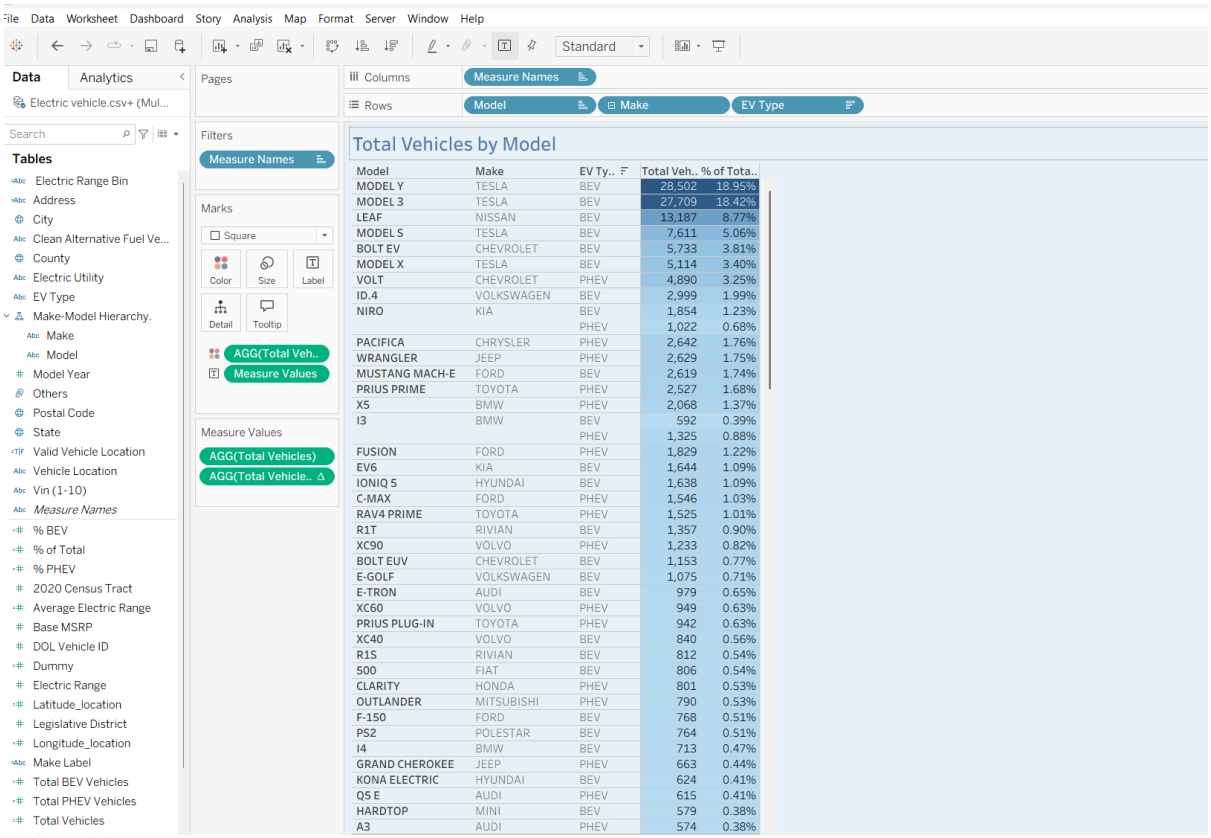


d) Total Vehicles by CAFV Eligibility:
Description: Illustrate the proportion of electric vehicles that are eligible for Clean Alternative Fuel Vehicle (CAFV) incentives, aiding in understanding the impact of incentives on vehicle adoption.



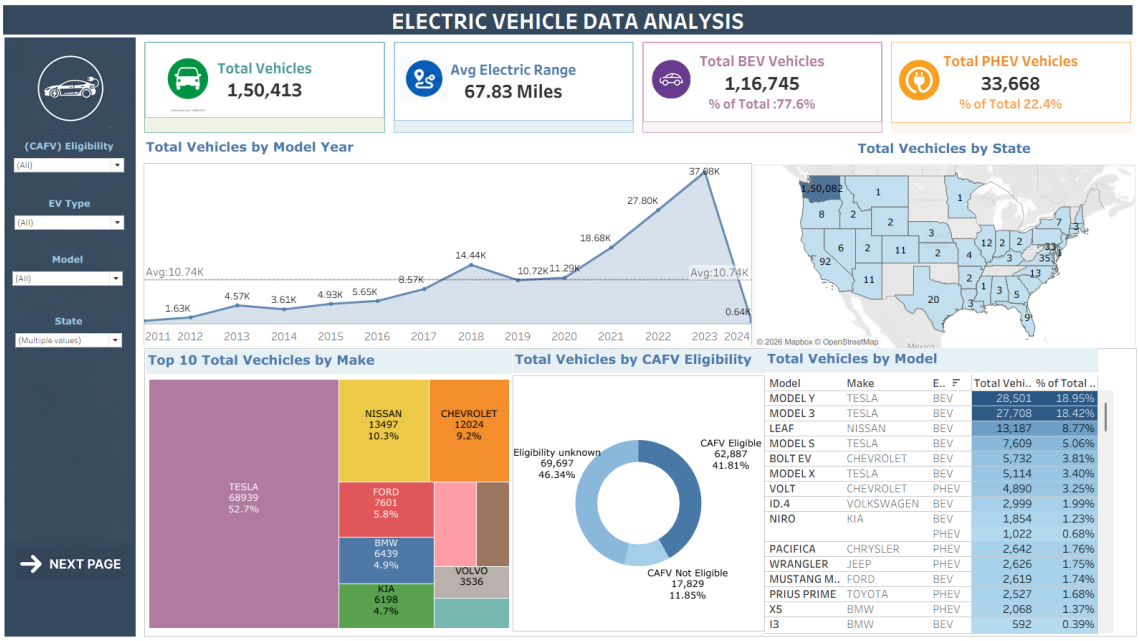
e) Top 10 Total Vehicles by Model:

Description: Highlight the top 10 electric vehicle models based on the total number of vehicles, offering insights into consumer preferences and popular models in the market.



Model	Make	EV Ty..	Total Veh..	% of Tota..
MODEL Y	TESLA	BEV	28,502	18.95%
MODEL 3	TESLA	BEV	27,709	18.42%
LEAF	NISSAN	BEV	13,187	8.77%
MODEL S	TESLA	BEV	7,611	5.06%
BOLT EV	CHEVROLET	BEV	5,733	3.81%
MODEL X	TESLA	BEV	5,114	3.40%
VOLT	CHEVROLET	PHEV	4,890	3.25%
ID.4	VOLKSWAGEN	BEV	2,999	1.99%
NIRO	KIA	BEV	1,854	1.23%
		PHEV	1,022	0.68%
PACIFICA	CHRYSLER	PHEV	2,642	1.76%
WRANGLER	JEEP	PHEV	2,629	1.75%
MUSTANG MACH-E	FORD	BEV	2,619	1.74%
PRIUS PRIME	TOYOTA	PHEV	2,527	1.68%
X5	BMW	PHEV	2,068	1.37%
I3	BMW	BEV	592	0.39%
		PHEV	1,325	0.88%
FUSION	FORD	PHEV	1,829	1.22%
EV6	KIA	BEV	1,644	1.09%
IONIQ 5	HYUNDAI	BEV	1,638	1.09%
C-MAX	FORD	PHEV	1,546	1.03%
RAV4 PRIME	TOYOTA	PHEV	1,525	1.01%
R1T	RIVIAN	BEV	1,357	0.90%
XC90	VOLVO	PHEV	1,233	0.82%
BOLT EUV	CHEVROLET	BEV	1,153	0.77%
E-GOLF	VOLKSWAGEN	BEV	1,075	0.71%
E-TRON	AUDI	BEV	979	0.65%
XC60	VOLVO	PHEV	949	0.63%
PRIUS PLUG-IN	TOYOTA	PHEV	942	0.63%
XC40	VOLVO	BEV	840	0.56%
R1S	RIVIAN	BEV	812	0.54%
500	FIAT	BEV	806	0.54%
CLARITY	HONDA	PHEV	801	0.53%
OUTLANDER	MITSUBISHI	PHEV	790	0.53%
F-150	FORD	BEV	768	0.51%
PS2	POLESTAR	BEV	764	0.51%
I4	BMW	BEV	713	0.47%
GRAND CHEROKEE	JEEP	PHEV	663	0.44%
KONA ELECTRIC	HYUNDAI	BEV	624	0.41%
Q5 E	AUDI	PHEV	615	0.41%
HARDTOP	MINI	BEV	579	0.38%
A3	AUDI	PHEV	574	0.38%

5. Below is the sample Dashboard view which can be taken as a sample. **DO NOT copy this dashboard exactly** but take the idea from this and use your own Analytical skills and creativity to create an end to end dashboard. Please add 2 Navigation buttons also in this dashboard , so that when user clicks on that button it will take him to the other screen/chart/dashboard. On that other screen/chart/dashboard, add 1 Navigation button with the "Back" option. So that when user clicks on "Back" button, user will come back to main dashboard.



6. Create the story also, starting with the dashboard and then moving towards into more granular level.

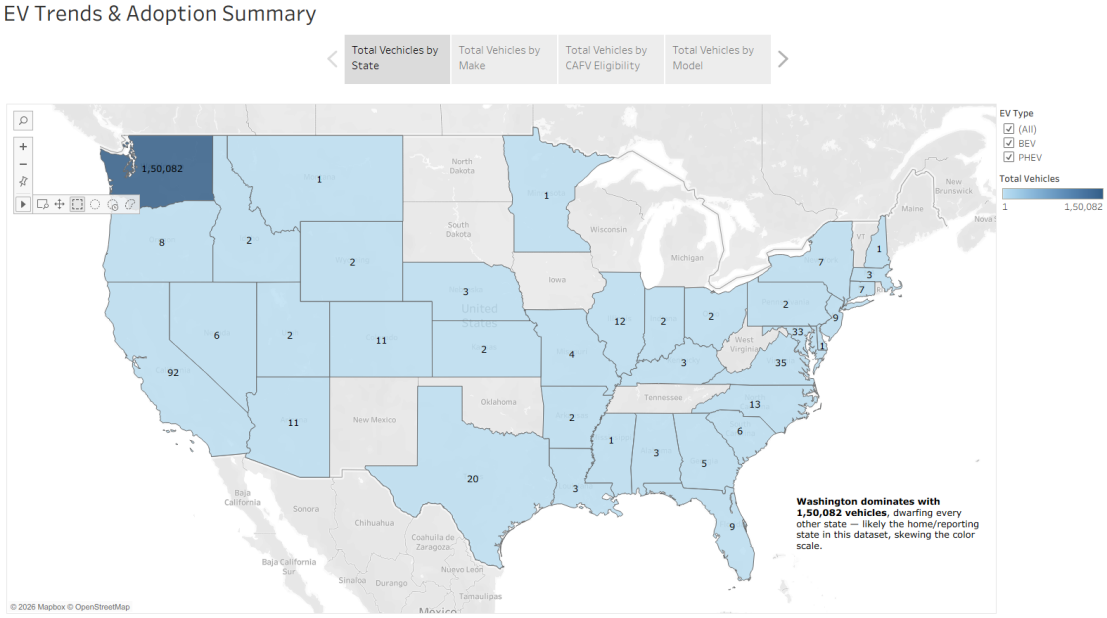


Tableau Link:-

[EV Vehicle Data Analysis | Tableau Public]